



Electrochemical passive properties of $\text{Al}_x\text{CoCrFeNi}$ ($x = 0, 0.25, 0.50, 1.00$) alloys in sulfuric acids

Yih-Farn Kao^a, Tsung-Dar Lee^a, Swe-Kai Chen^{b,*}, Yee-Shyi Chang^a

^a Department of Materials Science and Engineering, National Tsing Hua University, 101 Kuang Fu Road, Sec. 2, Hsinchu 30013, Taiwan, ROC

^b Center for Nanotechnology, Materials Science, and Microsystems, National Tsing Hua University, 101 Kuang Fu Road, Sec. 2, Hsinchu 30013, Taiwan, ROC

ARTICLE INFO

Article history:

Received 27 July 2009

Accepted 18 November 2009

Available online 27 November 2009

Keywords:

A. Alloy

B. EIS

B. Polarization

B. Weight loss

C. Passive films

ABSTRACT

This study investigates the electrochemical passive properties of $\text{Al}_x\text{CoCrFeNi}$ alloys in H_2SO_4 by potentiodynamic polarization, EIS, and weight loss tests from 20 to 65 °C. Experimental results indicate that Al harms the corrosion resistance in H_2SO_4 at temperatures exceeding 27 °C owing to the porous and inferior nature of the protection oxide film of Al in these alloys. Closely examining the Arrhenius plots of corrosion current density reveals that both pre-exponential factor A and activation energy E_a increase with Al content. However, A affects corrosion current density more significantly than E_a at higher temperatures and, conversely, at lower temperatures.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

As multiple principal element alloys, high-entropy alloys (HEAs) comprise at least five elements whose concentration for each one ranges between 5 and 35 at. % [1]. Attributes of forming a simple solid solution and nano particle precipitation, as well as achieving a high hardness and strength, and excellent high-temperature oxidation resistance make HEAs highly promising for application and research and development of these alloys [2–5]. Properties of $\text{Al}_x\text{CoCrFeNi}$ ($0 \leq x \leq 1$) HEAs vary significantly with x [6]. For instance, the alloy structure changes from FCC to BCC for increased Al content x . Additionally, the coefficient of thermal expansion decreases with x . Both properties are closely related to the bond strength of alloys. Moreover, electrical resistivity of $\text{Al}_x\text{CoCrFeNi}$ alloys is large, i.e., approximately up to 200 $\mu\Omega$ cm [7].

Corrosion properties of $\text{AlCoCrCu}_{0.5}\text{FeNiSi}$ [8,9], $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ [10,11], and $\text{Al}_{0.5}\text{CoCrCuFeNiB}_x$ [12] HEAs have been extensively studied in recent years. Among these HEAs, $\text{AlCoCrCu}_{0.5}\text{FeNiSi}$ alloy (HEA 1) displays, at room temperature, a better general corrosion resistance than SS 304 in 1 N H_2SO_4 ; however, it exhibits a worse pitting corrosion resistance than SS 304 in 1 N H_2SO_4 and in 1 M NaCl, respectively. The general corrosion resistance of each of HEA 1 and SS 304 decreases when exceeding room temperature. The effect of temperature on corrosion resistance of

HEA 1 is less severe in 1 M NaCl than in 1 N H_2SO_4 [9]. $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ alloys (HEA 2) reveal that in each of the 0.5 M H_2SO_4 and 1 M NaCl solutions, corrosion resistance increases with a decreasing x ; in addition, the susceptibility to general and pitting corrosion of HEA 2 increases with an increasing x [10]. $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ alloys (called hereinafter as HEA 2a and 2b for $x = 0$ and 0.3, respectively) in 0.1 M HCl exhibit different corrosion behaviours for different x values. Although HEA 2a is susceptible to localized corrosion, HEA 2b has a stable passive film on the surface. In 0.1 M HCl, anodized treatment of HEA 2a and 2b alloys in 15% H_2SO_4 gives higher corrosion resistance than the untreated [11]. In deaerated 1 N H_2SO_4 , $\text{Al}_{0.5}\text{CoCrCuFeNiB}_x$ alloys are more resistant to general corrosion than SS 304, and are not susceptible to localized corrosion. Additionally, the corrosion resistance of $\text{Al}_{0.5}\text{CoCrCuFeNiB}_{0.6}$ alloy is inferior to $\text{Al}_{0.5}\text{CoCrCuFeNi}$ alloy [12]. Above HEAs show an extremely close compositional dependence of corrosion behaviour in various solutions.

Although many interesting topics have been explored for $\text{Al}_x\text{CoCrFeNi}$ alloys [6,7], investigation on their corrosion property is still lacking. Therefore, this study elucidates how Al affects their corrosion behaviour. The electrochemical properties of the alloys in sulfuric acids are investigated using the potentiodynamic polarization and a weight loss measurement method. Additionally, these alloys are compared with SS 304, especially with respect to the effect of temperature. Moreover, based on use of electrochemical impedance spectroscopy (EIS), the effect of Al on corrosion behaviour is analysed. Furthermore, the relationship of stability of oxide film with Al content is examined by varying the chloride concentration in a sulfuric solution.

* Corresponding author. Tel.: +886 3 574 2569; fax: +886 3 571 3113.

E-mail address: skchen@mx.nthu.edu.tw (S.-K. Chen).

2. Experimental details

2.1. Test specimens for electrochemical tests and weight loss measurements

As-cast $\text{Al}_x\text{CoCrFeNi}$ alloys were prepared according to molar ratios of $x = 0, 0.25, 0.50$, and 1.00 (called C-0, C-0.25, C-0.50, and C-1.00, respectively) in a vacuum arc remelter. Table 1 lists the composition of the alloys. Test specimens were cut in $0.8 \text{ cm} \times 0.8 \text{ cm} \times 0.3 \text{ cm}$ and cold-mounted in epoxy with the outside surface from a surface of $0.8 \text{ cm} \times 0.8 \text{ cm}$ of specimens. The specimens were subsequently ground and polished with silicon carbide paper, rinsed and dried in preparation for electrochemical tests and weight loss measurements. During determination of the weight loss, six sets of samples were dipped in sulfuric acid for 1, 3, 5, 8, 11, and 15 days, respectively. All tests, except the weight loss test, were performed at least three times to confirm the data reproducibility. Finally, weight loss tests were performed twice and the reproducibility was given in an error bar.

2.2. Test solutions

The base solution for the test was 0.5 M of sulfuric acid. Test temperatures were ambient temperature ($\sim 25^\circ\text{C}$). Test solutions bearing chloride ions were with $0.25, 0.50$, and 1.00 M sodium chloride in the base solution. To avoid the dissolved oxygen (aeration) affecting the test solutions, deaeration was simultaneously made by a nitrogen gas flow of 120 ml/min in the test solution. The effect of temperature on polarization was examined under thermostatic control at an interval of 15°C in the temperature range of 20 – 65°C .

2.3. Electrochemical test

A three-electrode cell was used for the electrochemical test. The reference electrode was a commercial Ag/AgCl electrode saturated in 3 M KCl electrode ($-0.205 \text{ V}_{\text{SHE}}$ or -0.205 V to standard hydrogen electrode). The auxiliary electrode was made of Pt , and the working electrode was the specimen. Potentiostat was CH Instrument Model-600A. The specimen was cathodically polarized at a potential of $-0.4 \text{ V}_{\text{SHE}}$ for 300 s before the test for the purpose of removing surface oxides. The quasi-steady-state time for an open circuit was 900 s . Scan speed was 1 mV/s for scan potential ranging from -0.6 to $1.4 \text{ V}_{\text{SHE}}$. For EIS, the working potential was that of open circuit at 900 s from the start of immersion with scan amplitude 10 mV and a frequency ranging from 100 kHz to 10 mHz .

2.4. Scanning electron microscopy (SEM) metallograph examination and energy dispersed X-ray spectroscopy (EDS) analysis

Samples were fine polished, up to $0.05 \mu\text{m}$ Al_2O_3 powder and, then, examined with SEM (JEOL JSM-840A) equipped with an Oxford EDS for topography and elemental compositions. Finally, samples were examined before and after 3-day immersion of 0.5 M H_2SO_4 .

Table 1
Composition (wt.%) for alloys C- x and SS 304.

Alloys	Al	Co	Cr	Fe	Ni
C-0	0	27.12	23.74	23.99	25.14
C-0.25	3.05	25.14	22.48	24.15	25.18
C-0.50	5.59	25.25	22.13	22.80	24.22
C-1.00	10.02	23.84	21.11	21.99	23.03
SS 304	0	0	19.40	72.68	7.92

3. Results and discussion

3.1. Potentiodynamic polarization and weight loss at 25°C

Fig. 1 shows the anodic dissolution behaviour of alloys in 0.5 M H_2SO_4 , while Table 2 summarizes relevant data. This figure reveals a well-defined passive region of 0 – $1.2 \text{ V}_{\text{SHE}}$ in all curves. All curves, except for the one at $x = 0.25$ (C-0.25), show a secondary passive region at $0.15 \text{ V}_{\text{SHE}}$. This passivation is attributed mainly to the further oxidation or hydroxidation of the passive oxide film, thus altering the valence of Cr [13,14]. Fig. 1 also indicates that the secondary passive regions of C-0.50 and C-1.00 are more prominent than those of C-0 and C-0.25. This observation is due to the selective dissolution in the duplex FCC–BCC structure for C-0.50 and in the BCC-ordered BCC structure for C-1.00, as compared with C-0 and C-0.25 which are single FCC phase. In the active–passive transition region, different compositions at different secondary passivation potentials reveal different dissolution rates owing to a selective dissolution. This observation resembles that observed in duplex phase stainless steel [15,16].

Table 2 reveals that the corrosion potential (E_{corr}) and the corrosion current density (I_{corr}) for all of the alloys differ only slightly, and no obvious trends occur for E_{corr} and I_{corr} vs. x variation. The above phenomenon can be attributed to the spontaneous passivation of pure Al in H_2SO_4 [17]. Al metal spontaneously passivates in H_2SO_4 , explaining why its corrosion potential is ready in the passive region, i.e., this passivation explains why the polarization curve of Al does not display an apparent active–passive transition region. However, elements such as Cr and Fe exhibit a large critical current density (I_{cri}) for passivation, explaining why Cr and Fe dissolve more than Al before the alloy reaches its passive state. Thus, the variation of Al affects the active region of the polarization curves only slightly. Furthermore, in H_2SO_4 , all Al , Co , Cr , Fe , and Ni metals show passivity. Among them, Al has a relatively high passive current density (I_{pass}) [17,18] because only Al oxide easily forms porous film on the metal surface [19]. Therefore, protection by oxide layer on the alloys with higher Al content is inferior to that with lower Al content. Fig. 1 thus reveals that I_{pass} increases with x .

The results of potentiodynamic polarization were compared via performing 15-day-dipping and weight loss experiments. In the 15-day-dipping and weight loss experiments, the corrosion rates for C-0.50 and C-1.00 were markedly higher than those of C-0 and C-0.25 (Fig. 2). This observation differs substantially from the values of I_{corr} obtained from polarization experiment (Fig. 1), in which the two groups differ only slightly, despite the fact that the trend is the same. A previous study found a similar deviation

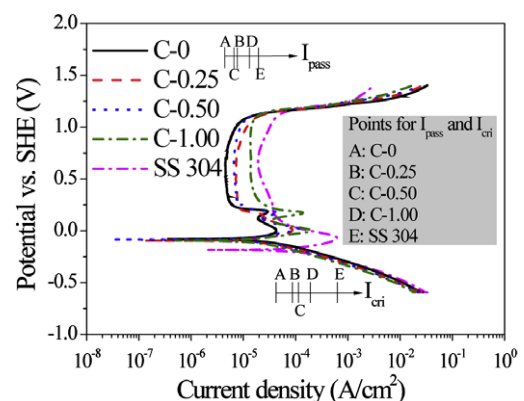


Fig. 1. Polarization diagrams for alloys C- x and SS 304 at 25°C .

Table 2
Polarization diagram parameters of alloys C-x and SS 304 at 25 °C.

Alloys C-x and SS 304	E_{corr} (V _{SHE})	I_{corr} (μA/cm ²)	E_{pp} (V _{SHE})	I_{crit} (μA/cm ²)	I_{pass} (μA/cm ²)
C-0	−0.081	15.8	0.002	42.8	4.5
C-0.25	−0.095	16.7	0.008	87.4	7.1
C-0.50	−0.084	13.4	0.017	117.2	6.4
C-1.00	−0.094	13.1	0.010	198.0	13.9
SS 304	−0.185	45.3	−0.071	603.0	19.1

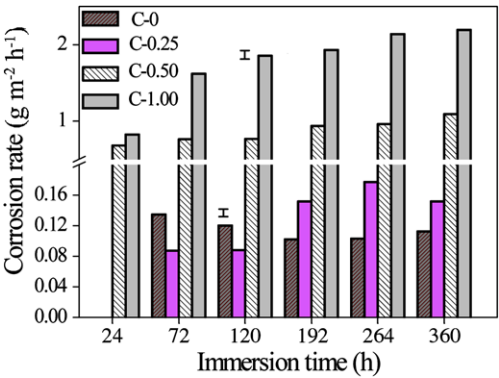


Fig. 2. Diagram showing change in corrosion rate (g m^{−2} h^{−1}) in the 15-day-dipping and weight loss measurement for alloys C-x.

in corrosion current densities obtained from weight loss test and potentiodynamic polarization method [20].

Fig. 1 shows polarization diagrams for the Al_xCoCrFeNi alloys and SS 304. The alloys have better overall general corrosion behaviour, with a larger E_{corr} and smaller I_{corr} , I_{crit} , and I_{pass} than SS 304.

3.2. Effect of temperature on potentiodynamic polarization

Fig. 3 shows polarization diagrams of Al_xCoCrFeNi alloys at various temperatures. A rising temperature decreased the Tafel slopes of anode (Table 3), increased I_{corr} , and increased E_{corr} and E_t (the transpassive potential) slightly. The corrosion rate is directly related to I_{corr} , according to Arrhenius equation, $I_{\text{corr}} = A \exp(-E_a/RT)$ [9,21], where the pre-exponential factor A is generally independent of temperature and is a constant of alloys, where R denotes the gas constant, T denotes temperature, and E_a denotes activation energy for corrosion. In the case of small experimental temperature range, E_a is assumed to be independent of T . Consequently, E_a can be obtained from $\ln(I_{\text{corr}})$ vs. $1/T$ plot. Fig. 4 shows such plots for the alloys and SS 304, indicating that E_a increases with x . This finding suggests that the corrosion rate is more sensitive to temperature for a larger Al content than for a smaller Al content. The $\ln(I_{\text{corr}})$ vs. $1/T$ curves intersect with each other in a range of 23–27 °C. Beyond this temperature range, I_{corr} increases with x . The situation is reversed at temperatures lower than 23 °C, which is inconsistent with a situation in which E_a s for all alloys increase with x from

Table 3
Fit data for Tafel slopes of alloys C-x in 20–65 °C.

Alloys C-x	20 °C		35 °C		50 °C		65 °C	
	β_a^a	β_c^b	β_a	β_c	β_a	β_c	β_a	β_c
C-0	158	−218	128	−158	134	−162	89	−158
C-0.25	158	−178	103	−167	89	−168	92	−149
C-0.50	94	−158	113	−178	138	−159	89	−198
C-1.00	104	−148	93	−173	98	−242	100	−220

^a Anodic Tafel slope β_a in mV/decade, the measured Tafel regions are with 40–50 mV of overvoltage.

^b Cathodic Tafel slope β_c in mV/decade, the measured Tafel regions are with 150–170 mV of overvoltage.

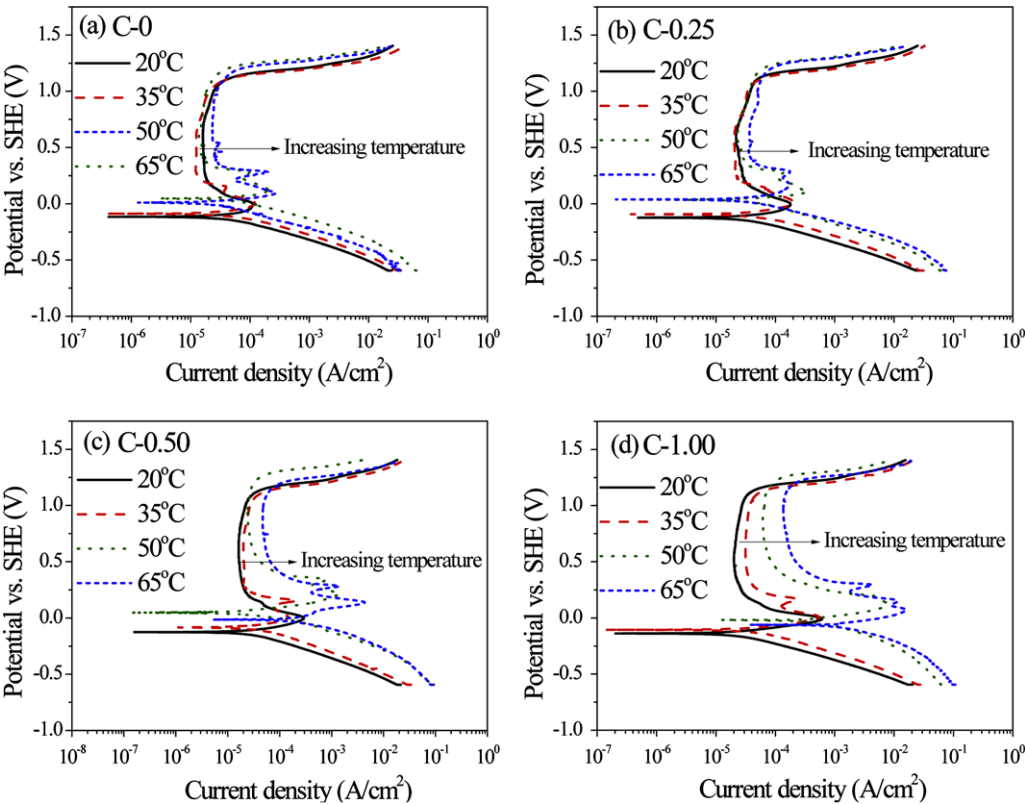


Fig. 3. Polarization diagrams for (a) C-0, (b) C-0.25, (c) C-0.50, and (d) C-1.00 at various temperatures.

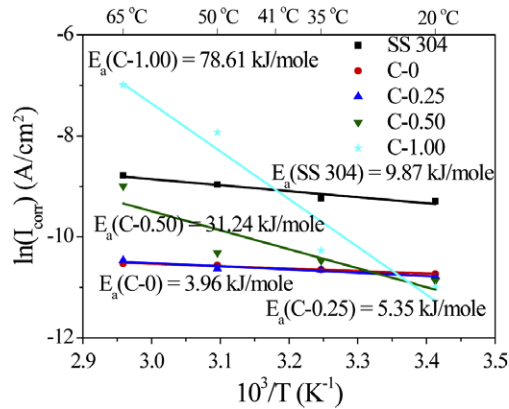


Fig. 4. The Arrhenius plots for alloys C-x and SS 304 at 20–65 °C.

20 to 65 °C. Hence, E_a , i.e., an intrinsic property of metal, and A , i.e., a surface property of metal, are determinative factors of I_{corr} . While E_a depends only on x , A depends on both x and temperature (Table 4). Therefore, although E_a increases with x , A also increases with x . Combining the effects of E_a and A explains the different corrosion behaviours of the alloys with an increasing x at temperatures exceeding 27 °C and lower than 23 °C. Thus, the performance of passive films, when Al is added, at higher temperatures becomes inferior to that without addition of Al. In determining I_{corr} , A is more important than E_a at temperatures exceeding 27 °C, while E_a is more important than A at temperatures lower than 23 °C.

3.3. EIS test at 25 °C

Figs. 5 and 6 summarize the EIS results of alloys in a sulfuric solution and their schematic equivalent circuit diagrams, respectively. Table 3 lists related parameters of the equivalent circuit diagrams, where R_s , R_f , and R_{ct} denote impedances of the sulfuric solution, oxide layer, and adsorption layer, Q_f and Q_{ad} denote capacitances of constant phase element (CPE) for oxide layer and adsorption layer, and L_{ad} denotes the inductance of the adsorption layer, respectively. Next, the oxide layer thickness is evaluated by using the Helmholtz model [22] and expressing the layer thickness of the oxide layer, d , as $d = \epsilon \epsilon_0 S / Q_f$, where ϵ_0 denotes the permittivity of free space (8.85×10^{-14} F/cm), ϵ denotes the dielectric constant of the medium, and S denotes the surface area of the electrode. Assuming that ϵ and S for all oxide layers of alloys are the same allows us to compare relative values of d for all samples by $1/Q_f$. Fig. 7 reveals that $1/Q_f$ values are proportional to x , implying that d increases with Al content x . However, according to this figure, the impedance of oxide layer R_f decreases with x and, in Fig. 8, the impedance of the oxide layer is inversely proportional to I_{pass} . Restated, a thinner oxide layer implies a larger value of impedance. To explain this phenomenon, besides the thickness of oxide layer, the density of oxide layer is also considered. As mentioned in Section 3.1, Al oxide easily forms a porous film on the me-

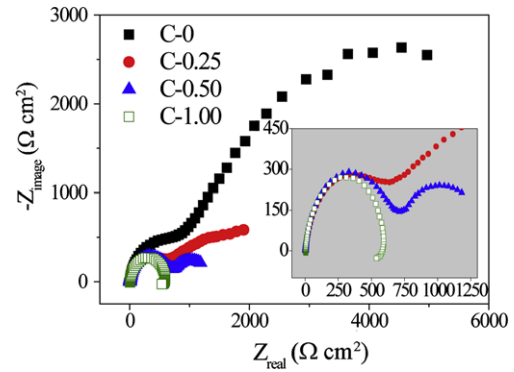


Fig. 5. The Nyquist plots for alloys C-x at 25 °C.

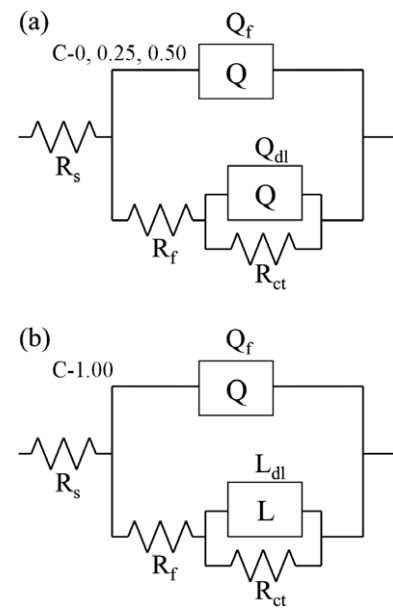


Fig. 6. EIS equivalent circuits (a) for alloys C-0, C-0.25, and C-0.50, and (b) for alloy C-1.00.

tal surface [19]. Therefore, it is easily understood that in addition to causing a thicker oxide layer, Al element promotes the dispersive oxide layer. Combining these two effects obviously reveals that R_f decreases with x .

According to Fig. 6, only C-1.00 reveals a component of inductance in the equivalent circuit. In previous studies [23–26], alloys with Al and Ni readily react with $(\text{OH})^-$ and $(\text{SO}_4)^{2-}$ in a sulfuric solution and adsorbed on the surface of the alloys, which increases the amount of the ions in the adsorption layer. Therefore, Q_{ad} increases with x , as listed in Table 5. As x value increases to 1.00, the inductance appears in the equivalent circuit in Fig. 6(b). This

Table 4

The fits for $A(x, T)$ and $E_a(x)$ in $I_{\text{corr}}(x, T) = A(x, T) \exp(-E_a(x)/RT)$.

Alloys	$A(x, T)$ (A/cm^2)				E_a (kJ/mol)
	20 °C (293 K)	35 °C (303 K)	50 °C (323 K)	65 °C (338 K)	
C-0	1.16×10^{-4}	1.16×10^{-4}	1.16×10^{-4}	1.07×10^{-4}	3.96
C-0.25	1.90×10^{-4}	2.03×10^{-4}	1.77×10^{-4}	1.90×10^{-4}	5.35
C-0.50	7.17	5.64	3.70	8.41	31.24
C-1.00	1.78×10^9	7.46×10^8	1.89×10^9	1.31×10^9	78.61
SS 304	1.18×10^{-4}	1.28×10^{-4}	1.70×10^{-4}	2.07×10^{-4}	9.87

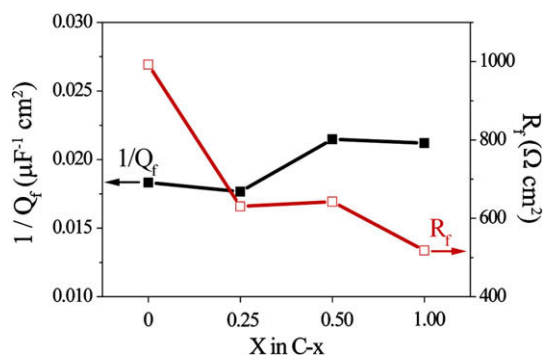


Fig. 7. Impedance and relative thickness ($1/Q_f$) of oxide layer vs. Al content x plots.

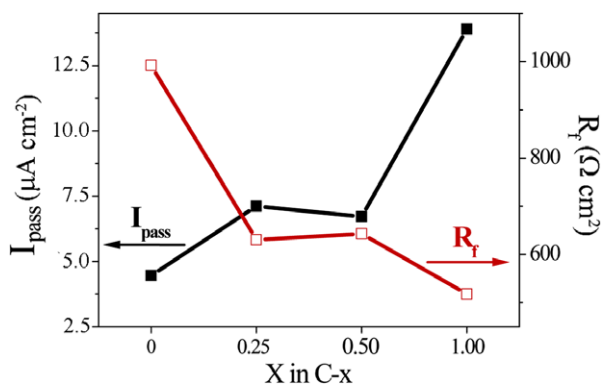


Fig. 8. Impedance and I_{pass} of oxide layer vs. Al content x plots.

effect normally occurs in the case of a severe corrosive condition [27]. Origin of the inductance can generally be influenced by some adsorbed intermediates or can be attributed to a space at the interfaces [28]. In C-1.00 with an Al and Ni-rich phase in a microstructure, which is seen as a reactive phase from metallograph, not only causes adsorption in these Al and Ni-rich areas in corrosion process, but also decreases the impedance in the low frequency area owing to their continuous dissolution. The fact that R_{ct} decreases with x demonstrates a higher dissolution rate for alloys with a higher Al content.

3.4. Polarization behaviour for alloys in a chloride-containing H_2SO_4 solution

Fig. 9 shows polarization diagrams for the alloys in 0.5 M H_2SO_4 solution containing various concentrations of chloride ions, as well as in simple 0.5 M H_2SO_4 solution as a comparison. According to Fig. 9(a), oscillation occurs in a passive region for C-0 in 0.5 M H_2SO_4 containing 0.5 and 1 M of chloride. This phenomenon has been attributed to the cycling process for small pitting and re-passivation with the duration of several seconds for each cycle [29]. Oscillation in the passive region in potentiodynamic polarization curve is a metastable state [30]. This metastable state generally re-

flects the difficulty of pitting, i.e., alloy C-0 has good anti-pitting ability, while those containing aluminum (C-0.25, C-0.50, and C-1.00) with no metastable state show an inferior anti-pitting ability (Fig. 9(b)–(d)).

From an adsorption viewpoint, adsorption competition always prevails on the alloy surface between chloride ions and dissolved oxygen atoms. Notably, no oxide layer forms once chloride ions adsorb on the alloy surface, in which the metal ions readily dissolve. Therefore, the adsorption of chloride ions increases the reacting current density (as indicated by a comparison of Figs. 1 and 9), subsequently increasing the rate of metal dissolution.

Rapid dissolution of alloys in chloride-containing solution is discussed next. When chloride ions are adsorbed on the interface of passive layer and a sulfuric solution, metastable ion complexes gradually form from the anions of a passive layer. These metastable ion complexes enable the anions to dissolve. Once the ion complexes on the passive layer/solution interface dissolve into the sulfuric solution, the inner ion complexes of the passive layer move to the passive layer/solution interface in order to correlate with the applied potential. The inability of the anions to form oxide implies the continuous formation of metastable ion complexes and dissolution of ions. Since Al easily forms $[\text{Al}(\text{SO}_4)]^+$ with $(\text{SO}_4)^{2-}$, and $\text{Al}(\text{OH})\text{SO}_4$ with $(\text{SO}_4)^{2-}$ and $(\text{OH})^-$, respectively [31], these metastable ion complexes combine with Cl^- and dissolve afterwards. Therefore, pitting easily occurs on the surface of aluminum alloys. Next, the aluminiferous passive layer and non-aluminiferous passive layer are compared. Fig. 10 shows the pitting potential (E_{pit}) of the alloys and SS 304 in different solutions. The value of E_{pit} for C-0 is almost independent of chloride concentration. The value of E_{pit} for C-0.25 decreases abruptly for a chloride concentration exceeding 0.50 M. This value is close to that of SS 304. The values of E_{pit} for C-0.25, C-0.50, and C-1.00 decrease to 0.2–0.5 V_{SHE} at a chloride concentration of 0.25 M (Fig. 10). A higher Al concentration in the alloys implies a lower value of E_{pit} . For C-0, deterioration of the passive layer is attributed to the evolution of oxygen. Meanwhile, for C-0.25, C-0.50, and C-1.00, the deterioration of passive layer is attributed to the pitting process. An increasing chloride ion concentration causes the chloride ions to cluster at the defect sites of the passive layer and severely attack the passive layer. Consequently, E_{pit} shifts to a more active region.

3.5. Metallographic examination and EDS analysis

Microstructures for not H_2SO_4 -immersed alloys C-0, C-0.25, C-0.50, and C-1.00 are with single FCC, single FCC, duplex FCC–BCC, and BCC-ordered BCC phases, respectively [6]. Table 6 lists the EDS composition for each phase in different alloys.

Fig. 11(a) and (b) show the microstructure of C-0 before and after 3-day immersion in 0.5 M H_2SO_4 , respectively. Fig. 11(c) and (d) show the microstructure of C-0.25 before and after 3-day immersion in 0.5 M H_2SO_4 , respectively. General corrosion occurs for both C-0 and C-0.25, as revealed by EDS analyses (see the bold values shown in Table 6).

Fig. 12(a) and (b) show the microstructure of C-0.50 before and after 3-day immersion in 0.5 M H_2SO_4 , respectively. According to

Table 5
EIS equivalent circuit parameters for alloys C- x .

Alloys C- x	R_s ($\Omega \text{ cm}^2$)	Q_f ($\mu\text{F}/\text{cm}^2$)	n_f	R_f ($\Omega \text{ cm}^2$)	Q_{ad} ($\mu\text{F}/\text{cm}^2$)	n_{ad}	R_{ct} ($\Omega \text{ cm}^2$)
C-0	3.271	54.57	0.9094	992.2	636.7	0.7444	7691
C-0.25	3.758	56.61	0.9081	610.5	1525	0.6347	1932
C-0.50	2.994	46.55	0.9223	642.8	3221	0.6454	819.1
C-1.00	3.462	47.16	0.9614	518.1	122.4 (H) ^a	–	66.81

^a $L_{\text{ad}} = 122.4$ Henry.

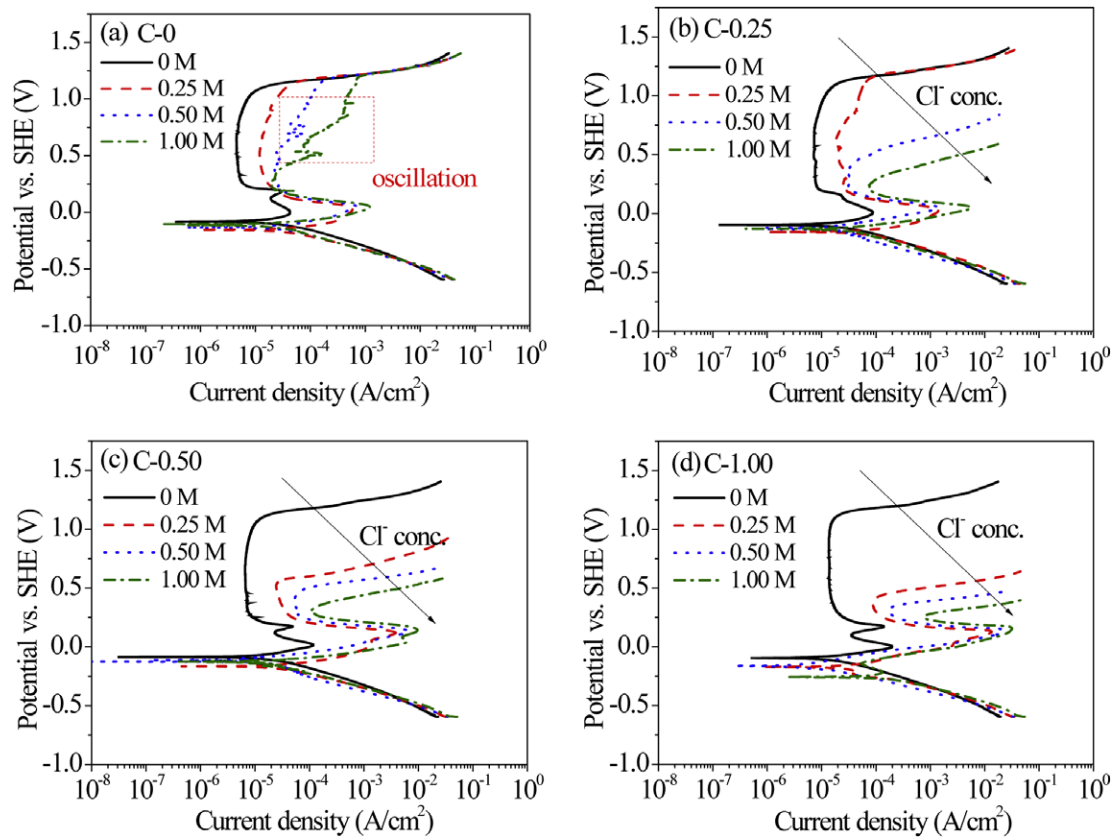


Fig. 9. Polarization diagrams for (a) C-0, (b) C-0.25, (c) C-0.50, and (d) C-1.00 at 25 °C in chloride-containing sulfuric acid solution at various Cl^- molarity (M) values.

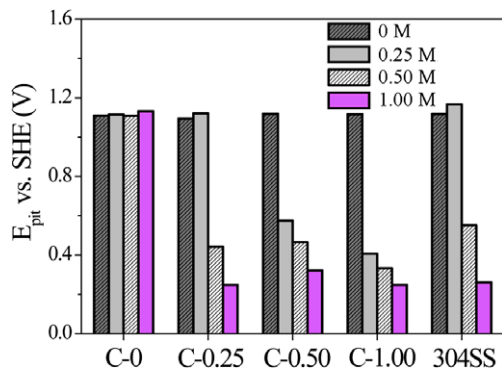


Fig. 10. Histogram of E_{pit} for alloys C-x and SS 304 in solution of different Cl^- ion molarity (M).

these figures, after immersion the FCC phase remains smooth while the BCC phase shows a rough morphology. Fig. 12(c) shows a line-scanned area across the FCC and BCC phases for an immersed sample. Fig. 12(d) summarizes the line-scanned results, indicating that the BCC phase of C-0.50 before immersion is rich in Al and Ni. However, after immersion, it is poor in Al and Ni and rich in Cr.

Fig. 13 shows the microstructure and line-scan analysis of C-1.00 before and after immersion. Before immersion, BCC and ordered BCC phases cannot be resolved from the microstructure. The composition of BCC phase after alloy immersion is close to the overall alloy composition before immersion, indicating that the BCC phase is a corrosion-resistant phase. Moreover, the change in overall composition after immersion is attributed to the selective dissolution of Al and Ni in the ordered BCC phase of this alloy (see the bold values shown in Table 6).

Table 6

EDS analyses (at. %) for alloys C-0, C-0.25, C-0.50, and C-1.00.

Alloys	Phases and states	Al	Co	Cr	Fe	Ni
C-0	Overall, not immersed	0	25.93	25.73	24.21	24.13
	Overall, immersed	0	24.45	26.39	24.83	24.32
C-0.25	Overall, not immersed	6.16	23.27	23.58	23.59	23.40
	Overall, immersed	6.18	23.65	24.41	23.04	22.71
C-0.50	Overall, not immersed	11.01	22.77	22.61	21.70	21.92
	FCC matrix, not immersed	8.36	24.74	23.48	22.77	20.65
	BCC, not immersed	13.94	21.11	20.48	20.53	23.94
	Overall, immersed	8.35	22.75	27.19	23.60	18.11
	FCC matrix, immersed	9.96	22.57	23.16	23.59	20.72
	Wall-shaped BCC, immersed	3.82	22.50	36.33	25.58	11.77
C-1.00	Overall, not immersed	18.88	20.55	20.63	20.01	19.93
	Overall, immersed	12.45	19.80	29.87	23.42	14.45
	BCC, immersed	17.14	20.67	21.96	20.89	19.34
	Ordered BCC, immersed	3.04	17.34	47.53	27.96	4.14

This selective corrosion in Al and Ni-rich phase in C-0.50 and C-1.00, which results in the corrosion attack on Al and Ni, is due to the large bonding in Al and Ni [32]. Alloys containing this bonding readily react with $(\text{OH})^-$ and $(\text{SO}_4)^{2-}$ to form Al and Ni complexes and dissolve in a sulfuric solution. Accordingly, after immersion, the remaining compound in the less corrosive-resistant Al and Ni-rich phase is an oxide, rich in Cr, in the residual passive film.

4. Conclusions

Owing to the spontaneous passivation of Al element in H_2SO_4 , the variation of Al reveals a more apparent effect in a passive region rather than in an active one. Therefore, in contrast with I_{pass} ,

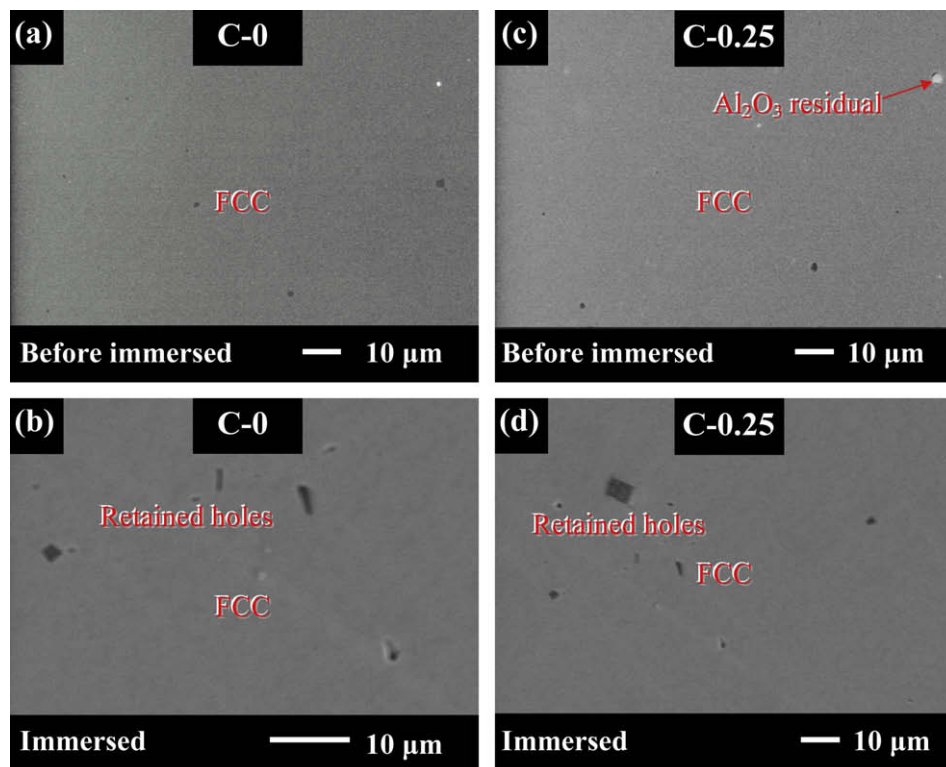


Fig. 11. Metallograph of alloys C-0 (a and b) and C-0.25 (c and d). (a) and (c), before; and (b) and (d), after immersion. Retained holes were from cast procedure, and Al₂O₃ residuals were from polishing procedure.

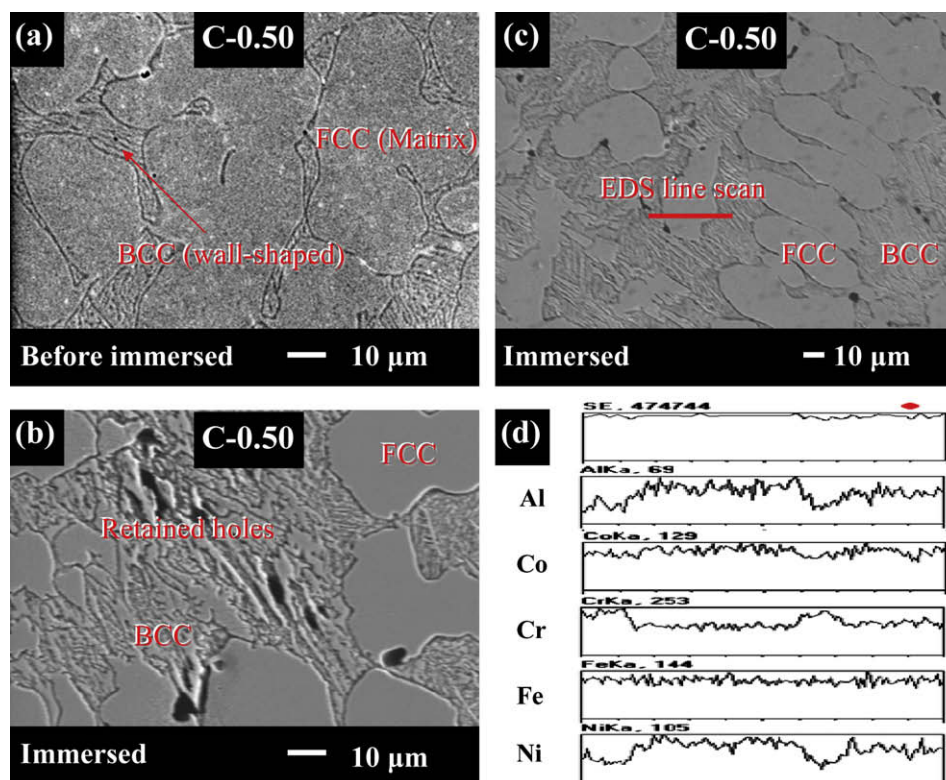


Fig. 12. Metallograph of alloy C-0.50. (a), before; (b) and (c), after immersion; and (d), EDS line-scan results of the location indicated in (c). Retained holes were from cast procedure.

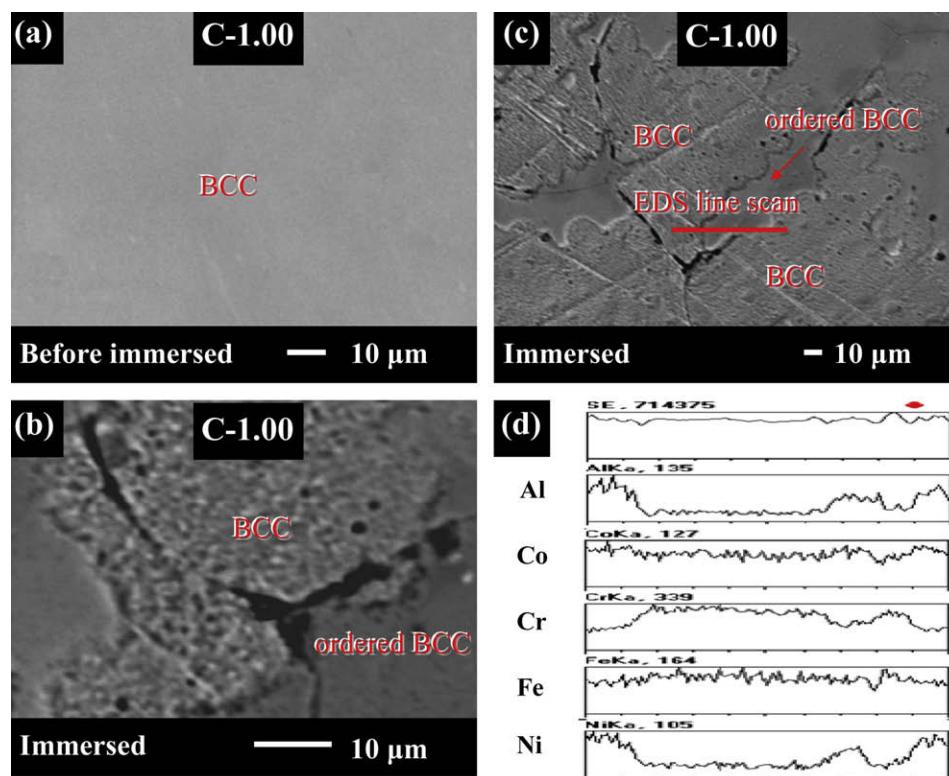


Fig. 13. Metallograph of alloy C-1.00. (a), before; (b) and (c), after immersion; and (d), EDS line-scan results of the location indicated in (c).

which increases with x , no obvious trends occur for E_{corr} and I_{corr} vs. x variation. In particular, the weight loss experiment indicates that I_{pass} is a proper index to evaluate the weight loss of samples since $\text{Al}_x\text{CoCrFeNi}$ alloys are found to have passive behaviour in long-term dipping.

EIS results indicate that the passive films of $\text{Al}_x\text{CoCrFeNi}$ alloys become increasingly thicker and more dispersive with an increasing x . Therefore, I_{pass} increases with x . As x value increases to 1.00, the inductance effect appears in the equivalent circuit for severe dissolution of Al and Ni-rich phase. As for the effect of chloride on the anti-corrosion property, chloride eases the passive layer to form metastable ion complexes and further dissolve into H_2SO_4 . With an increasing chloride concentration and Al content, the metastable ion complexes easily form, allowing E_{pit} to shift to a more active region. Additionally, the microstructure of both C-0 and C-0.25 is single FCC phase, while those of C-0.50 and C-1.00 are duplex FCC–BCC and complex BCC–ordered BCC phase, respectively. In C-0.50 and C-1.00, the secondary passivation phenomenon in polarization curve results from selective dissolution of the Al and Ni-rich phase.

Moreover, I_{corr} increases with x at higher temperatures ($>27^\circ\text{C}$), while I_{corr} decreases with x at lower ones ($<23^\circ\text{C}$). More closely examining Arrhenius plots of I_{corr} reveals that both pre-exponential factor A and activation energy E_a increase with Al content. However, A affects I_{corr} more significantly than it does so for E_a at higher temperatures ($>27^\circ\text{C}$) and, conversely, at lower temperatures ($<23^\circ\text{C}$).

Acknowledgements

The authors would like to thank the National Science Council of the Republic of China, Taiwan (Contract No. NSC96-2221-E007-066-MY3) and the CSC Group Education Foundation of China Steel Corporation for financially supporting this research. Ted Knoy is

appreciated for his editorial assistance. Professor Tung Hsu is acknowledged for his aid in English of this paper.

References

- [1] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes, *Adv. Eng. Mater.* 6 (2004) 299–303.
- [2] P.K. Huang, J.W. Yeh, T.T. Shun, S.K. Chen, Multi-principal-element alloys with improved oxidation and wear resistance for thermal spray coating, *Adv. Eng. Mater.* 6 (2004) 74–78.
- [3] C.Y. Hsu, J.W. Yeh, S.K. Chen, T.T. Shun, Wear resistance and high-temperature compression strength of FCC $\text{CuCoNiCrAl}_{0.5}$ Fe alloy with boron addition, *Metall. Mater. Trans. A* 35A (2004) 1465–1469.
- [4] J. Tong, S.K. Chen, J.W. Yeh, T.T. Shun, C.H. Tsau, S.J. Lin, S.Y. Chang, Mechanical performance of the $\text{Al}_x\text{CoCrCuFeNi}$ high-entropy alloy system with multiprincipal elements, *Metall. Mater. Trans. A* 36A (2005) 1263–1271.
- [5] J.W. Yeh, S.K. Chen, J.Y. Gan, S.J. Lin, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Formation of simple crystal structures in Cu–Co–Ni–Cr–Al–Fe–Ti–V alloys with multiprincipal metallic elements, *Metall. Mater. Trans. A* 35A (2004) 2533–2536.
- [6] Y.F. Kao, T.J. Chen, S.K. Chen, J.W. Yeh, Microstructure and mechanical property of as-cast, -homogenized, -deformed $\text{Al}_x\text{CoCrFeNi}$ ($0 \leq x \leq 2$) high-entropy alloys, *J. Alloys Compd.* (2009), doi:10.1016/j.jallcom.2009.08.090.
- [7] H.P. Chou, Y.S. Chang, S.K. Chen, J.W. Yeh, Microstructure, thermophysical and electrical properties in $\text{Al}_x\text{CoCrFeNi}$ ($0 \leq x \leq 2$) high-entropy alloys, *Mater. Sci. Eng. B* 163 (2009) 184–189.
- [8] Y.Y. Chen, T. Duval, U.D. Hung, J.W. Yeh, H.C. Shih, Microstructure and electrochemical properties of high entropy alloys—a comparison with type-304 stainless steel, *Corros. Sci.* 47 (2005) 2257–2279.
- [9] Y.Y. Chen, U.T. Hong, H.C. Shih, J.W. Yeh, T. Duval, Electrochemical kinetics of the high entropy alloys in aqueous environments—a comparison with type 304 stainless steel, *Corros. Sci.* 47 (2005) 2679–2699.
- [10] C.P. Lee, C.C. Chang, Y.Y. Chen, J.W. Yeh, H.C. Shih, Effect of the aluminium content of $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ high-entropy alloys on the corrosion behaviour in aqueous environments, *Corros. Sci.* 50 (2008) 2053–2060.
- [11] C.P. Lee, Y.Y. Chen, C.Y. Hsu, J.W. Yeh, H.C. Shih, Enhancing pitting corrosion resistance of $\text{Al}_x\text{CrFe}_{1.5}\text{MnNi}_{0.5}$ high-entropy alloys by anodic treatment in sulfuric acid, *Thin Solid Films* 517 (2008) 1301–1305.
- [12] C.P. Lee, Y.Y. Chen, C.Y. Hsu, J.W. Yeh, H.C. Shih, The effect of boron on the corrosion resistance of the high-entropy alloys $\text{Al}_{0.5}\text{CoCrCuFeNiB}_x$, *J. Electrochem. Soc.* 154 (2007) C424–C430.
- [13] V. Ashworth, P.J. Boden, Potential–pH diagrams at elevated temperatures, *Corros. Sci.* 10 (1970) 709–718.

- [14] Y.Y. Chen, L.B. Chou, L.H. Wang, J.C. Oung, H.C. Shih, Electrochemical polarization and stress corrosion cracking of alloy 690 in 5-M chloride solutions at 25 °C, *Corrosion* 61 (2005) 3–11.
- [15] M. Femenia, J. Pan, C. Laygraf, In situ local dissolution of duplex stainless steels in 1 M H₂SO₄ + 1 M NaCl by electrochemical scanning tunneling microscopy, *J. Electrochem. Soc.* 149 (2002) B187–B197.
- [16] I.H. Lo, W.T. Tsai, Effect of selective dissolution on fatigue crack initiation in 2205 duplex stainless steel, *Corros. Sci.* 49 (2007) 1847–1861.
- [17] F.D. Bogar, M.H. Peterson, A comparison of actual and estimated long-term corrosion rate of mild steel in seawater, *Laboratory Corrosion Test and Standards*, ASTM STP 866 (1985) 197–206.
- [18] P. Marcus, *Corrosion Mechanisms in Theory and Practice*, second ed., Marcel Dekker, New York, 2002.
- [19] V. Shankar Rao, V.S. Raja, Anodic polarization and surface composition of Fe–16Al–0.14C alloy in 0.25 M sulfuric acid, *Corrosion* 59 (2003) 575–583.
- [20] L. Young, *Anodic Oxide Films*, first ed., Academic Press, London, 1961.
- [21] G.K. Gomma, Corrosion of low-carbon steel in sulphuric acid solution in presence of pyrazole–halides mixture, *Mater. Chem. Phys.* 55 (1998) 241–246.
- [22] C.F. Zinola, A.M. Castro Luna, The inhibition of Ni corrosion in H₂SO₄ solutions containing simple non-saturated substances, *Corros. Sci.* 37 (1995) 1919–1929.
- [23] M.R.F. Hurtado, P.T.A. Sumodjo, A.V. Benedetti, Electrochemical studies with a Cu–5 wt.%Ni alloy in 0.5 M H₂SO₄, *Electrochim. Acta* 48 (2003) 2791–2798.
- [24] F. M Reis, H.G. de Melo, I. Costa, EIS investigation on Al 5052 alloy surface preparation for self-assembling monolayer, *Electrochim. Acta* 51 (2006) 1780–1788.
- [25] T.M. Yue, L.J. Yan, C.P. Chan, C.F. Dong, H.C. Man, G.K.H. Pang, Excimer laser surface treatment of aluminum alloy AA7075 to improve corrosion resistance, *Surf. Coat. Technol.* 179 (2004) 158–164.
- [26] I. Epelboin, C. Gabrielle, M. Keddam, H. Takenouti, Achievements and tasks of electrochemical engineering, *Electrochim. Acta* 22 (1975) 913–920.
- [27] M. Metikoš-Huković, R. Babić, S. Brinić, EIS-in situ characterization of anodic films on antimony and lead–antimony alloys, *J. Power Sources* 157 (2006) 563–570.
- [28] A.R. Trueman, Determining the probability of stable pit initiation on aluminium alloys using potentiostatic electrochemical measurements, *Corros. Sci.* 47 (2005) 2240–2256.
- [29] Y.M. Tang, Y. Zuo, X.H. Zhao, The metastable pitting behaviours of mild steel in bicarbonate and nitrite solutions containing Cl[−], *Corros. Sci.* 50 (2008) 989–994.
- [30] R.T. Foley, T.H. Nguyen, The chemical nature of aluminum corrosion, *J. Electrochem. Soc.* 129 (1982) 464–467.
- [31] S. Van Gils, C.A. Melendres, H. Terryn, E. Stijns, Use of in-situ spectroscopic ellipsometry to study aluminium/oxide surface modifications in chloride and sulfuric solutions, *Thin Solid Films* 455 (2004) 742–746.
- [32] H. Nakazawa, H. Sato, Bacterial leaching of cobalt-rich ferromanganese crusts, *Int. J. Miner. Process.* 43 (1995) 255–265.